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Educational Control Products

Manual For Model 505

Inverted Pendulum

(Instructor's Edition)

Important Notice

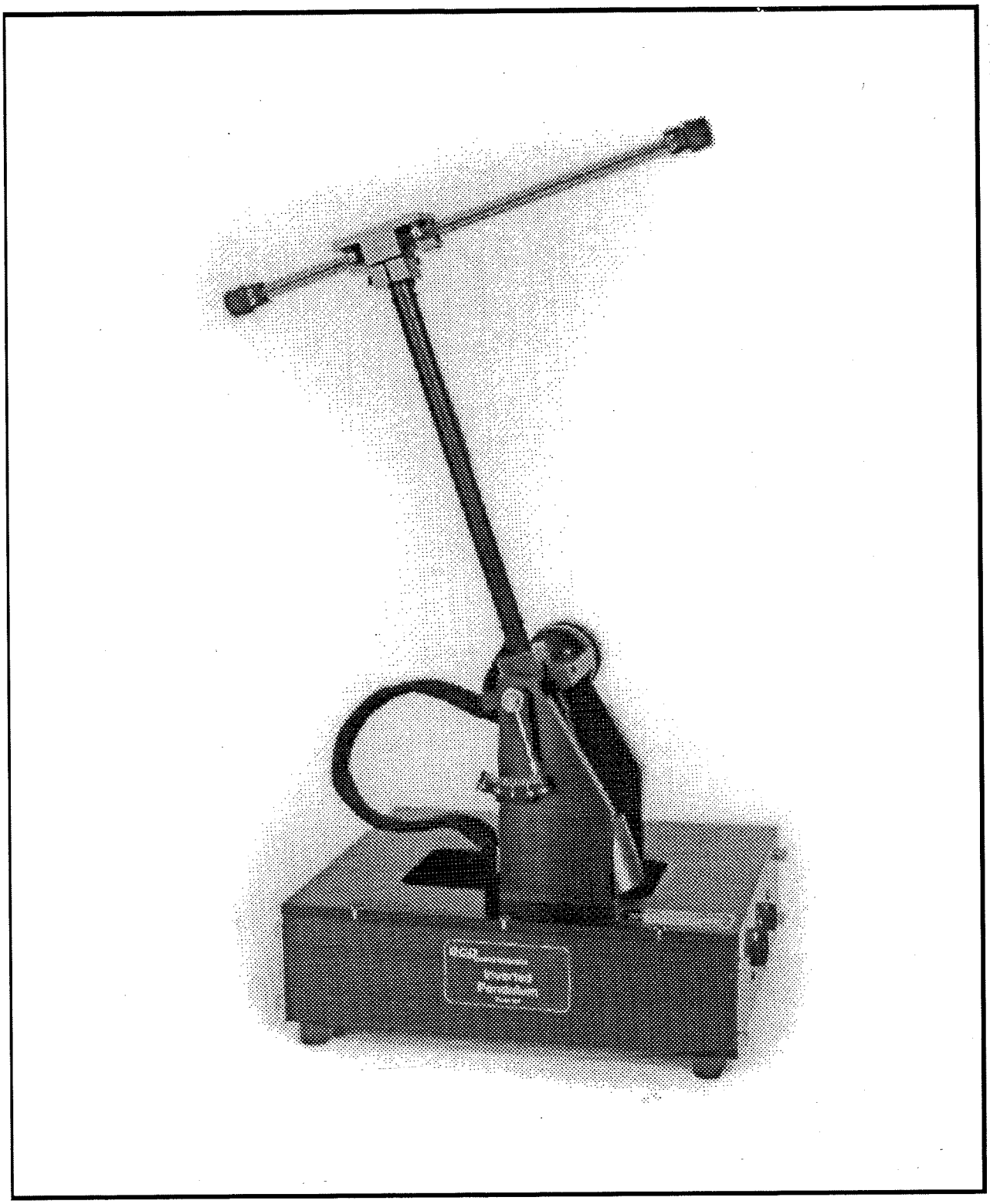
Section 2.3 of this manual contains important safety information that must be read by all users prior to operating the ECP system.

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This Edition is compatible with Executive Softwarte Version 2.0

ecp
Educational Control Products
1 Buckskin Court
Bell Canyon, CA 91307
ph: (818) 703-0802



The Inverted Pendulum

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Appendix Ai. Useful Scripts

Listed in this appendix are Matlab® scripts and functions for building the numerical plant models; designing the PD / pole placement and LQR controllers; and nonlinear simulation from Chapter 6. These are not represented as being numerically or methodologically optimal, but may be useful to some users.

A.1i Plant Model Builder

```
%Penplant.m
% Initialization file for numerical value of plant & control hardware
parameters
% Also includes a set of vectors and matrices for the lqr design of state
feedback controller
% This .m file must be executed prior to running the function lq(...).

%% INITIALIZE
% The following user-specified values vary with plant configuration
mw1=.110;    % Mass of Donut weights, set = zero if weights not used (kg)
mw2=1.00;    % Balance mass (kg)
lw=-0.1084; % Signed distance from pivot to c.g. of balance mass (m)

% Misc data required for other scripts - Change As Appropriate
ts=0.00442; % Sampling time (s)
d01=[0;0;0;0];          %Initial disturbance used in nonlsim function for
simulation only
% Vectors and matrices used for lqr design
R=[2 5 10 20 50]; % control weighting for lqr design used in lq function
C=[1,0,0,0];      % C vector in state-space model (output is angle)
Q=C'*C;           % Q matrix for lqr design used in lq function

% Fixed values:
g=9.807; %(m/s^2)
lo=0.330;    %pendulum rod length (m)
lco= 0.0711;       %lc when mw2=0 (m)
m1o=.103;    % Mass of sliding rod without donut weights (kg)
m2o=0.785;   % Mass of pendulum assembly minus m1 and minus balance mass (kg)
Jos=.0246;         %Polar moment of inertia about the pivot of pend ass'y minus
sliding rod and balance weight (kg-m^2)
% Scale factors of hardware (sensors and actuator and digital controller)
kx=50200;    % Sliding rod scale factor (encoder counts/m)
ka=2546;           % Angle encoder scale factor (encoder counts/radian)
kf=0.0013;   % Combined DAC amplifier scale factor (N/increment of DAC)
ks=32;       % Controller card firmware scale factor (increments/increments)

% Derived values
m1=m1o+mw1;
m2=m2o+mw2;
```

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```
lc=(m2o*lco+mw2*lw)/m2;        %cg of the pendulum assembly minus the sliding
rod(m)
Joe=Jos+m1*lo^2+mw2*lw^2;      %Polar moment of inertia of complete pend ass'y
about the pivot (kg-m^2)
Js=Jos+mw2*lw^2;          %This is J*(=Joe-m1*lo^2) in the theory section (kg
m^2)


%% BUILD PLANTS
% Use The Following For Plant in The State Space Form:
% x = [angle; angular velocity; position; velocity]
%                      xdot = Ax + Bu;
%                         y = Cx
% For system in SI units use the following:
%a1=[0,1,0,0];
%a2=[m2*lc*g/Js,0,m1*g/Js,0];
%a3=[0,0,0,1];
%a4=[(Js-m2*lc*lo)*g/Js,0,-m1*lo*g/Js,0];
%B=[0;-lo;0;(Js+m1*lo^2)/m1]/Js;
%For prefilter gain, kpf, in lqr synthesis
%kpf1=-(m1*g);
%kpf2=-(m1*lo+m2*lc)/m1;

% For system in units of controller counts use...
a1=[0,1,0,0];
a2=[m2*lc*g/Js,0,m1*g/Js*ka/kx,0];
a3=[0,0,0,1];
a4=[(Js-m2*lc*lo)*g/Js*kx/ka,0,-m1*lo*g/Js,0];
B=[0;-lo*ka;0;(Js+m1*lo^2)/m1*kx]*ks*kf/Js;
%For kpf in lqr synthesis
kpf1=-(m1*g)/ka/kf/ks;
kpf2=-kx*(m1*lo+m2*lc)/m1/ka;

% The following are common to both forms
A=[a1;a2;a3;a4];
C=[1,0,0,0];       % Output vector (output is angle), use C=[0 0 1 0] for x(t)
output
%Use the following for linear quadratic synthesis
Q=C'*C;


% For The Transfer Function theta(s)/F(s) = Na(s)/D(s), Use:
% For Na in SI units use...
%Na =(lo/Js)*[-1 0 g/lo];
% For Na in units of controller counts use...
Na = ka*ks*kf*(lo/Js)*[-1 0 g/lo];
D = [1 0 (m1*lo-m2*lc)*g/Js 0 -m1*g^2/Js];

% For The Transfer Function x(s)/F(s) = Nx(s)/D(s), Use:
% For Nx in SI units use...
%Nx =(1/m1/Js)*[Joe 0 -(m1*lo+m2*lc)*g];
% For Nx in units of controller counts use...
Nx = kx*ks*kf*(1/m1/Js)*[Joe 0 -(m1*lo+m2*lc)*g];
D = [1 0 (m1*lo-m2*lc)*g/Js 0 -m1*g^2/Js];
```

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```
% For The Transfer Function theta(s)/x(s) = Nax(s)/Dax(s), Use:
% For Nax in SI units use...
% Nax=m1*lo/Joe*[-1 0 g/lo];
% For Nax in units of controller counts use...
Nax=ka/kx*m1*lo/Joe*[-1 0 g/lo];
Dax=[1 0 -(m1*lo+m2*lc)*g/Joe];
```

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